HTTP Requests and Responses

Beginner · 45 minutes · version 1

By the end of this lesson you will be able to:

- Explain GET and POST methods
- Distinguish between 2xx, 4xx, and 5xx response status categories

Intro and goal

Every time you load a webpage, your browser talks to a web server using HTTP messages.

Example - Imagine typing a web address into your browser and pressing enter. Behind the scenes, your browser sends a request, and the server replies with the page you wanted.

- Client Browser → HTTP Request: sends
- HTTP Request → Web Server: reaches
- Web Server → HTTP Response: returns
- HTTP Response → Client Browser: delivers

Visual: The basic client-server transaction cycle in HTTP.

- Explain GET and POST methods
- Distinguish between 2xx, 4xx, and 5xx response status categories

Overview of HTTP Protocol

HTTP is a stateless request-response protocol running in a client-server model.

Definition - Protocol: A set of rules defining how messages are formatted and exchanged between computers.

Definition - Client: The user agent, such as a web browser, that initiates requests.

Definition - Server: The host system that receives requests, processes them, and returns responses.

Definition - Stateless Protocol: A protocol where the server does not retain status information about each user across multiple requests.

- HTTP Protocol → Client: uses
- HTTP Protocol → Server: uses
- Client → Request: sends
- Server → Response: returns

Visual: Core relationships within the HTTP architecture.

Core idea 1: Understanding GET and POST Methods

Methods classify the desired action a client wants to perform on a resource.

Method	Primary Purpose	Cacheable?
GET	Retrieve a resource	Yes
POST	Submit data for processing	Yes

Visual: Comparison of GET and POST characteristics.

A request method is a verb in the start line of an HTTP request. GET is a safe method meant for read-only access. POST is used when you need to send data to the server.

Example - Using GET is like reading a book from a library shelf. Using POST is like filling out a paper form and handing it to the librarian to create a new library card.

- Library Shelf → GET Request: matches read-only action
- Submission Form → POST Request: matches data submission action

Visual: Analogy linking real-world actions to HTTP methods.

1. What term describes a request method like GET that has no intended effect on the server state?

Hint: Think about whether safe methods modify server data.

Core idea 2: Response Status Categories (2xx, 4xx, 5xx)

HTTP response status codes tell the client whether a request succeeded or failed, and who is responsible.

- Status Code Categories → 2xx Success: includes
- Status Code Categories → 4xx Client Error: includes
- Status Code Categories → 5xx Server Error: includes

Visual: The major response code categories.

A 2xx status indicates that the client request was successfully received, understood, and accepted. A 4xx status indicates an error caused by the client. A 5xx status indicates the server encountered an error.

Example - If you misspell a URL and get a 404, the client request had a problem. If the website crashes because its database is down, it returns a 500 server error.

Category	Meaning	Responsible Party
2xx	Success	None
4xx	Client Error	Client
5xx	Server Error	Server

Visual: Breakdown of HTTP response status categories.

2. Which HTTP status code category indicates an error caused by the client?

Hint: Look for the category starting with number 4.

Worked example: Analyzing an HTTP Transaction

Tracing a request and response reveals how methods and status codes work together.

```
Request Line
Response Line
```

Visual: Sample HTTP request and response start lines.

- Step 1: Client prepares a GET request for /customer/123. (Why: Retrieves a specific resource safely.)
- Step 2: Client sends the request over the established connection. (Why: Transmits the action verb and URI to the server.)
- Step 3: Server receives and processes the request. (Why: Determines if the resource exists and if the user is authorized.)
- Step 4: Server returns a response starting with HTTP/1.1 200 OK. (Why: Indicates success and that the requested resource entity is attached.)
- Step 5: If the server encountered a bug, it would return a 5xx status instead. (Why: Signals that the server failed to fulfill a valid request.)

- GET /customer/123 → Server Processing: triggers
- Server Processing → 200 OK Response: produces

Visual: Flow of the worked example transaction.

Practice: Try it: Identify Method and Status Category

Review a scenario where a client submits user profile updates using POST and receives a 500 error response. State which method was used and which status code category the error belongs to.

Test

This test consists of 10 questions covering GET/POST methods and status code categories. A hint is available for each question, and answers appear only after submission.

Objective	Tested Concepts	Question Count
LO-01	GET and POST methods	5
LO-02	Status code categories (2xx, 4xx, 5xx)	5

Visual: Test coverage overview.

3. Which HTTP method is defined as safe and intended for read-only resource retrieval?

- POST
- GET
- PUT
- DELETE

Hint: Think about retrieving versus submitting data.

4. What is the primary purpose of the POST method in HTTP?

- To delete existing server resources
- To submit data for server processing
- To retrieve a read-only document
- To check if a connection is alive

Hint: POST is associated with sending form data.

5. Which response status category signifies that a request was successful?

- 1xx
- 2xx
- 4xx
- 5xx

Hint: Think of the category that includes 200 OK.

6. An error status code caused by the client falls under which category?

- 2xx
- 3xx
- 4xx
- 5xx

Hint: Review who is responsible for 4xx errors.

7. If a web server encounters an internal error and cannot fulfill a valid request, what status category is returned?

- 2xx
- 3xx
- 4xx
- 5xx

Hint: Server faults are represented in the 5xx range.

8. A user tries to fetch a product catalog page using a safe, cacheable method. Which method did they use?

- POST
- GET
- PUT
- CONNECT

Hint: Look for the method defined as safe and read-only.

9. An application submits new user registration details to a web server. Which HTTP method is appropriate here?

- GET
- POST
- TRACE
- OPTIONS

Hint: Registration sends new data to the server.

10. A client requests a file, but the URL contains a typo resulting in a 404 Not Found. Who is responsible for this error?

- The client
- The server
- The proxy cache
- The transport layer

Hint: Review the responsibility definition for 4xx status codes.

11. Order the steps of a basic HTTP request-response transaction.

- Client sends HTTP request message → Server processes request
- Server processes request → Server returns HTTP response message

Visual: Transaction flow order.

Put these steps in order:

- Client sends HTTP request message
- Server processes request
- Server returns HTTP response message

Hint: Think about who initiates the transaction.

12. Match each HTTP status category to its description.

Match:

- 2xx = ____
- 4xx = ____
- 5xx = ____

Hint: Match category numbers to their outcome definitions.

Recap

You now understand how HTTP requests use methods like GET and POST, and how responses categorize outcomes using 2xx, 4xx, and 5xx codes.

- HTTP is a stateless request-response protocol between clients and servers.
- GET requests are safe, read-only operations used to retrieve resources.
- POST requests submit data to the server for processing.
- 2xx codes denote success, 4xx client errors, and 5xx server errors.

Concept	Key Summary
GET	Safe, read-only retrieval
POST	Submits data for action
2xx	Successful request
4xx	Client-side error
5xx	Server-side error

Visual: Cheat-sheet table for HTTP methods and response codes.

Next steps:

- Explore browser developer tools to inspect live HTTP traffic.
- Read about additional HTTP methods like PUT and DELETE.

Answer key

1. (Check) safe - GET is defined as a safe method meant for read-only access.

2. (Check) 4xx - 4xx status codes are designated for client-side errors.

3. (Test) GET - GET is a safe, read-only method.

4. (Test) To submit data for server processing - POST is used to send data to the server.

5. (Test) 2xx - 2xx status codes indicate success.

6. (Test) 4xx - 4xx codes indicate client-caused errors.

7. (Test) 5xx - 5xx codes indicate server-side errors.

8. (Test) GET - GET is safe and cacheable for fetching resources.

9. (Test) POST - POST is used for submitting data to be processed.

10. (Test) The client - 4xx errors are caused by the client request.

11. (Test) Client sends HTTP request message → Server processes request → Server returns HTTP response message - Transactions follow client request, server processing, and server response.

12. (Test) 2xx = Success; 4xx = Client error; 5xx = Server error - 2xx is success, 4xx is client error, 5xx is server error.

Videos and references

Video: The Http and the Web | Http Explained | Request-Response Cycle - https://www.youtube.com/watch?v=eesqK59rhGA
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